

TorchJD: Jacobian Descent in PyTorch

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github.com/SimplexLab/TorchJD



torchjd.org

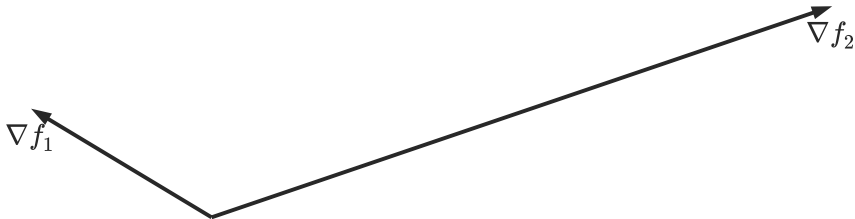


arxiv.org/pdf/2406.16232

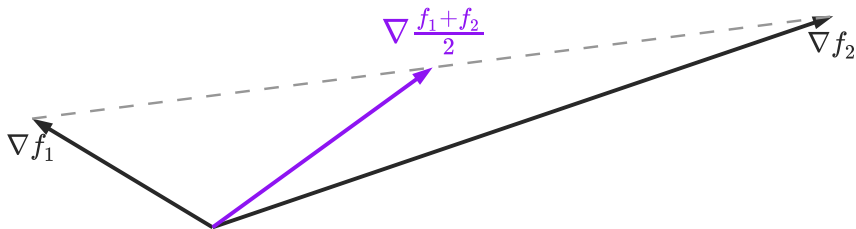
Motivation

Jacobian descent

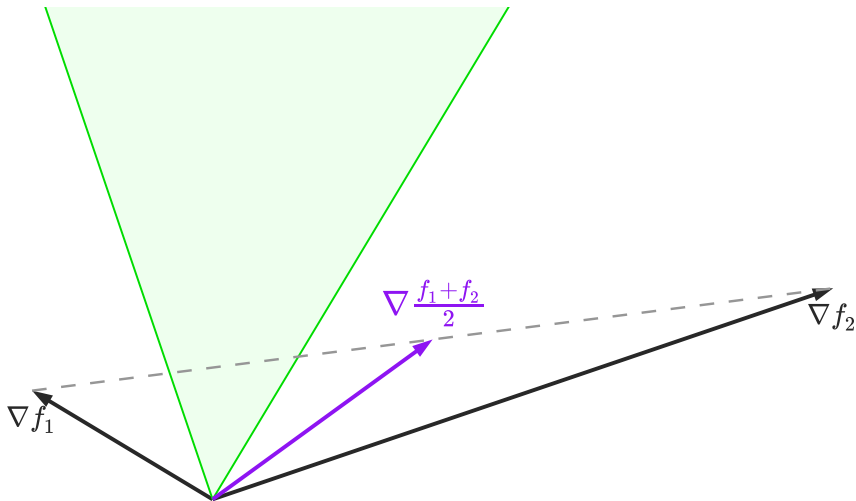
Jacobian descent



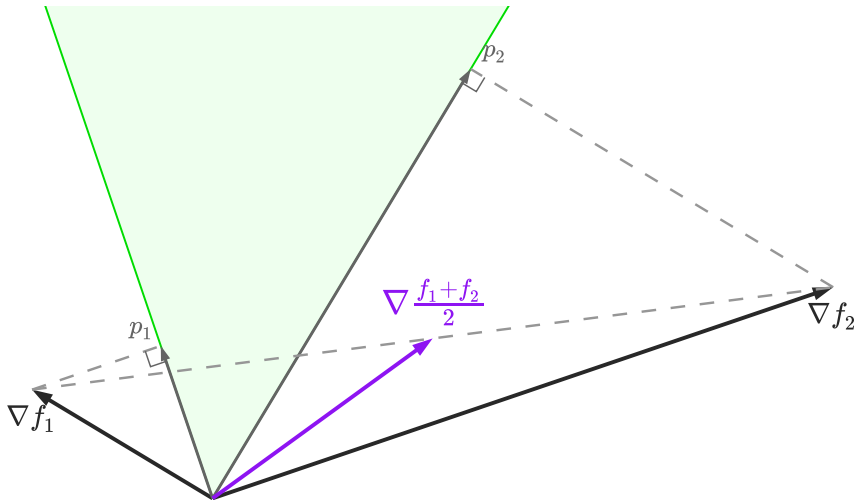
Jacobian descent



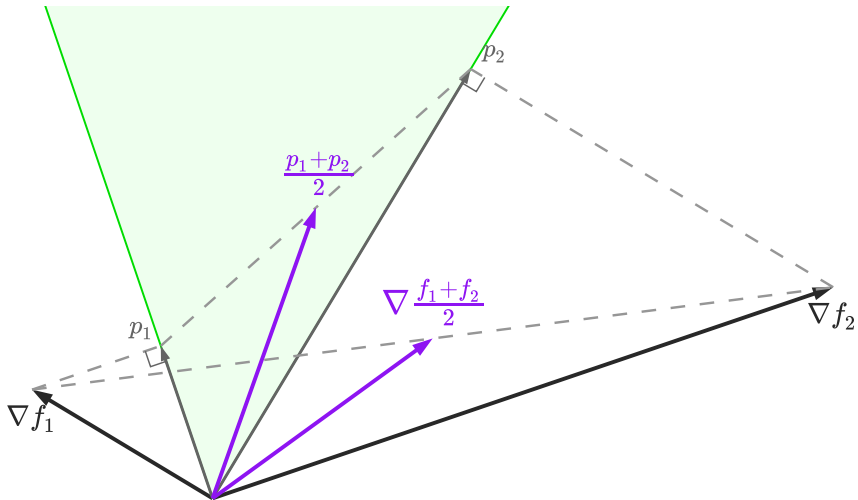
Jacobian descent



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Jacobian descent



TorchJD

TorchJD: core functions

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torchjd	autojac.backward	compute jacobians and accumulate in .jac
torchjd	autojac.jac_to_grad	aggregate .jac fields into .grad

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    autojac.jac_to_grad(model.parameters(), UPGrad())
    # parameters now have a .grad field
    optimizer.step()
    optimizer.zero_grad()
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Discussion